

Every prior, same slate

The held-out scores that decided which prior is served, then how each one would have ranked today's hitters.

Served (full season)

11
priors compared

32,641
held-out plate appearances

3.05%
held-out HR rate

0.5999
best AUC · KNN comps

served prior

Held-out scores

Fitted on plate appearances before 2026-08-27, scored on the 32,641 that came after (actual HR rate 3.05%). No outcome in the test window was visible during fitting. **Read AUC first** — the slate ranks hitters and takes the top six, so ordering is what matters. Brier and log loss are dominated by the ~3% base rate and barely separate the priors. Metrics are per plate appearance; the probabilities lower down are per game.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.5836	1.55×	0.02952	0.13556	0.00462
KNN comps Prior is the pooled rate of the nearest comparables.	0.5999	1.52×	0.02949	0.13463	0.00494
SVR Support-vector regression on the same feature matrix.	0.5969	1.48×	0.02953	0.13522	0.00618
Random forest Forest regression; a learned similarity metric.	0.5958	1.47×	0.02952	0.13504	0.00551
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.5999	1.52×	0.02949	0.13463	0.00494
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.5999	1.52×	0.02949	0.13463	0.00494
Linear regression Ordinary least squares on the same features.	0.5965	1.47×	0.02951	0.13498	0.00436
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5976	1.54×	0.02953	0.13512	0.00680
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5964	1.46×	0.02951	0.13496	0.00557
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5971	1.45×	0.02953	0.13503	0.00625

BLEND WEIGHTS (FOREST VARIANT)

KNN 1.00 SVR 0.00 RF 0.00

SHRINKAGE K PER PRIOR

league 248 knn 248 svr 248 rf 248 core 248 ensemble 248
linear 248 logistic 248 nn 248 xgb 248

Estimated by method of moments on out-of-fold residuals, not hand-tuned.

Thin-sample subset

The same held-out window, restricted to hitters with fewer than 100 plate appearances before the cutoff (3,547 PA, actual HR rate 2.34%). For a regular with 600 PA the observed record swamps any prior, so the priors mostly separate here.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.5526	1.44×	0.02288	0.11153	0.00903
KNN comps Prior is the pooled rate of the nearest comparables.	0.5744	1.20×	0.02296	0.11191	0.01517
SVR Support-vector regression on the same feature matrix.	0.5731	1.32×	0.02313	0.11507	0.01888
Random forest Forest regression; a learned similarity metric.	0.5821	1.08×	0.02314	0.11331	0.01866
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.5744	1.20×	0.02296	0.11191	0.01517
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.5744	1.20×	0.02296	0.11191	0.01517
Linear regression Ordinary least squares on the same features.	0.5768	0.60×	0.02307	0.11384	0.01888
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5769	0.48×	0.02314	0.11364	0.02055
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5743	1.32×	0.02303	0.11288	0.01760
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5886	1.08×	0.02315	0.11327	0.01986

Slate agreement

PRIOR	SHARED	ITS TOP SIX
Served (full season)	6/6	Matt Olson 17.0%, Ronald Acuña Jr. 16.4%, Fernando Tatis Jr. 15.8%, Shohei Ohtani 14.9%, Drake Baldwin 14.8%, Jackson Chourio 14.5%
Flat league	6/6	Fernando Tatis Jr. 15.7%, Ronald Acuña Jr. 15.7%, Matt Olson 15.4%, Drake Baldwin 14.5%, Jackson Chourio 14.5%, Shohei Ohtani 13.4%
KNN comps	6/6	Matt Olson 16.6%, Ronald Acuña Jr. 16.4%, Fernando Tatis Jr. 16.1%, Shohei Ohtani 15.0%, Drake Baldwin 15.0%, Jackson Chourio 14.9%
SVR	6/6	Matt Olson 17.2%, Ronald Acuña Jr. 16.5%, Fernando Tatis Jr. 15.9%, Drake Baldwin 15.2%, Shohei Ohtani 15.1%, Jackson Chourio 15.0%
Random forest	6/6	Matt Olson 17.0%, Ronald Acuña Jr. 16.3%, Fernando Tatis Jr. 15.7%, Drake Baldwin 15.6%, Shohei Ohtani 15.2%, Jackson Chourio 15.1%
Core (KNN+SVR)	6/6	Matt Olson 16.6%, Ronald Acuña Jr. 16.4%, Fernando Tatis Jr. 16.1%, Shohei Ohtani 15.0%, Drake Baldwin 15.0%, Jackson Chourio 14.9%
Forest blend	6/6	Matt Olson 16.6%, Ronald Acuña Jr. 16.4%, Fernando Tatis Jr. 16.1%, Shohei Ohtani 15.0%, Drake Baldwin 15.0%, Jackson Chourio 14.9%
Linear regression	6/6	Matt Olson 16.7%, Fernando Tatis Jr. 16.3%, Ronald Acuña Jr. 16.2%, Drake Baldwin 15.3%, Shohei Ohtani 14.9%, Jackson Chourio 14.9%
Logistic regression	6/6	Matt Olson 17.3%, Fernando Tatis Jr. 16.4%, Ronald Acuña Jr. 16.3%, Shohei Ohtani 15.3%, Drake Baldwin 15.1%, Jackson Chourio 14.9%
Neural net (torch)	6/6	Matt Olson 17.0%, Fernando Tatis Jr. 16.4%, Ronald Acuña Jr. 16.3%, Shohei Ohtani 15.2%, Drake Baldwin 15.0%, Jackson Chourio 15.0%
XGBoost	6/6	Matt Olson 17.0%, Ronald Acuña Jr. 16.2%, Fernando Tatis Jr. 15.7%, Drake Baldwin 15.6%, Shohei Ohtani 15.4%, Jackson Chourio 15.2%

Hitter by hitter

The top 40 bats by the served probability.

MLB Season Performance Analysis - Q3 2024											
HITTER	SERVED (FULL SEASON)	FLAT LEAGUE	KNN COMPS	SVR	RANDOM FOREST	CORE (KNN+SVR)	FOREST BLEND	LINEAR REGRESSION	LOGISTIC REGRESSION	NEURAL NET (TO	
1Matt Olson	slate	17.0%	15.4%	16.6%	17.2%	17.0%	16.6%	16.6%	16.7%	17.3%	
2Ronald Acuña Jr.	FL	16.4%	15.7%	16.4%	16.5%	16.3%	16.4%	16.4%	16.2%	16.3%	
3Fernando Tatis Jr.	slate	15.8%	15.7%	16.1%	15.9%	15.7%	16.1%	16.1%	16.3%	16.4%	
4Shohei Ohtani	slate	14.9%	13.4%	15.0%	15.1%	15.2%	15.0%	15.0%	14.9%	15.3%	
5Drake Baldwin	ATL	14.8%	14.5%	15.0%	15.2%	15.6%	15.0%	15.0%	15.3%	15.1%	
6Jackson Chourio	slate	14.5%	14.5%	14.9%	15.0%	15.1%	14.9%	14.9%	14.9%	14.9%	
7Jackson Merrill	SD	13.8%	12.4%	13.7%	13.7%	13.7%	13.7%	13.7%	13.7%	14.0%	
8Teoscar Hernández	SD	13.2%	12.5%	13.3%	13.4%	13.6%	13.3%	13.3%	13.5%	13.5%	
9Manny Machado	SD	13.1%	12.7%	13.2%	13.6%	13.3%	13.2%	13.2%	13.4%	13.5%	
10Will Smith	LAD	13.0%	12.7%	13.0%	13.1%	13.6%	13.0%	13.0%	13.6%	13.4%	
11William Contreras	AS	12.5%	12.3%	12.5%	12.3%	12.3%	12.5%	12.5%	12.4%	12.4%	
12Mookie Betts	LAD	12.1%	12.1%	11.9%	12.6%	11.9%	11.9%	11.9%	12.3%	12.3%	
13Jake Bauers	MIL	12.0%	11.2%	12.3%	12.2%	12.2%	12.3%	12.3%	12.1%	12.0%	
14Austin Riley	ATL	12.0%	11.0%	11.7%	11.7%	11.5%	11.7%	11.7%	11.7%	11.8%	
15Garrett Mitchell	IL	11.6%	11.1%	11.6%	10.2%	10.8%	11.6%	11.6%	11.0%	10.9%	
16Michael Harris II	FL	11.1%	11.2%	11.0%	11.6%	11.0%	11.0%	11.0%	11.3%	11.6%	
17Ty France	SD	10.8%	10.6%	11.0%	10.9%	11.1%	11.0%	11.0%	11.0%	11.0%	
18Brice Turang	MIL	10.7%	11.0%	10.5%	10.8%	10.4%	10.5%	10.5%	10.1%	10.1%	
19Andy Pages	LAD	10.0%	9.5%	9.6%	9.4%	9.7%	9.6%	9.6%	9.5%	9.4%	
20Sean Murphy	ATL	9.9%	9.5%	10.1%	9.4%	10.2%	10.1%	10.1%	10.3%	9.7%	
21Mike Yastrzemski	KL	9.7%	9.8%	9.9%	9.8%	9.7%	9.9%	9.9%	9.6%	9.5%	
22Ozzie Albies	ATL	9.7%	10.1%	9.6%	10.3%	10.3%	9.6%	9.6%	10.0%	9.9%	
23Freddie Freeman	LA	9.6%	9.3%	9.3%	9.2%	9.1%	9.3%	9.3%	9.4%	9.2%	
24Christian Yelich	MIL	9.4%	9.6%	9.7%	9.3%	9.6%	9.7%	9.7%	9.8%	9.4%	
25Xander Bogaerts	SD	9.0%	9.0%	8.8%	8.6%	8.8%	8.8%	8.8%	9.0%	8.7%	
26Kyle Tucker	LAD	9.0%	9.1%	9.1%	8.8%	9.0%	9.1%	9.1%	8.9%	8.7%	
27Luis Campusano	SD	7.7%	7.7%	7.9%	7.6%	7.8%	7.9%	7.9%	7.9%	7.6%	
28Joey Ortiz	MIL	7.3%	8.4%	7.5%	7.6%	7.5%	7.5%	7.5%	7.4%	7.4%	

HITTER	SERVED (FULL SEASON)	FLAT LEAGUE	KNN COMPS	SVR	RANDOM FOREST	CORE (KNN+SVR)	FOREST BLEND	LINEAR REGRESSION	LOGISTIC REGRESSION	NEURAL NET (TO
29Mauricio Dubón	7.1%	8.6%	7.2%	7.6%	7.4%	7.2%	7.2%	7.1%	7.4%	
30Dustin Harris	6.8%	8.2%	6.7%	5.8%	6.0%	6.7%	6.7%	6.6%	6.4%	
31Tommy Edman	6.1%	6.6%	6.0%	6.1%	5.6%	6.0%	6.0%	6.1%	6.0%	
32Ha-Seong Kim	5.7%	7.0%	5.0%	4.4%	4.3%	5.0%	5.0%	5.4%	5.3%	
33Cooper Pratt	5.4%	7.0%	5.4%	4.9%	5.1%	5.4%	5.4%	4.9%	4.9%	
34Sal Frelick	4.9%	6.0%	4.8%	4.6%	4.6%	4.8%	4.8%	4.7%	4.7%	
35Jake Cronenworth	4.7%	5.9%	4.8%	4.8%	4.8%	4.8%	4.8%	4.8%	4.8%	
36David Hamilton	4.3%	5.7%	4.5%	4.3%	4.0%	4.5%	4.5%	4.3%	4.3%	
37Luis Lara	4.3%	6.0%	4.6%	3.7%	4.0%	4.6%	4.6%	3.6%	4.2%	

Per-game probabilities: each prior's per-PA rate run through this hitter's own plate appearances, park, weather, bullpen and times-through-the-order terms. Only the prior changes across the columns — every other term is held fixed, so a left-to-right spread is the prior disagreeing about the hitter, not the context. Shading grows with the probability.